FOUR ACUTE FACES: AN ANTIPODAL COUNTING PROOF

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Abstract. Four independent uniform points on the unit sphere form a tetrahedron whose four faces are all acute with probability exactly 1/12. We give a self-contained exposition of the antipodal counting proof that first appeared in a conversation publicly shared by Timothy Chow. Antipodal flips turn the dependent angle conditions into a 4×4 table of vector sums whose marked entries form one of six corner shapes. Our contribution is the organisation: one table for all twelve angle tests, a counting identity expressed through the faces of a single tetrahedron, and a shorter final calculation using one spherical cap.

1. Antipodal flips

Choose four points independently and uniformly on the unit sphere in $\mathbb{R}^3$. What is the probability that their tetrahedron has four acute faces? A single face is acute with probability 1/2 [1, Lemma 4], but the faces are dependent, so multiplying four such probabilities does not give the answer.

Before trying to average twelve angle conditions, consider

$$a = \tfrac{1}{3}(-2,-1,2), \quad b = (1,0,0), \quad c = \tfrac{1}{3}(-2,1,2), \quad d = \tfrac{1}{3}(-1,2,2). \tag{1}$$

Consider the angle at c in acd, before and after replacing c by $-c$. The first dot product is $(a-c)\cdot(d-c) = -2/9$. After the replacement, the three angle dot products in $a(-c)d$ are 4/9, 28/9 and 2/3. The face has become acute. In fact $(a,-b,-c,d)$ has four acute faces; the table below will verify all twelve angles at once. For these points, exactly one of the eight tetrahedra up to overall reflection succeeds.

The question was posed on MathOverflow and Mathematics Stack Exchange [5, 6]. On 14 August 2026, Timothy Chow linked a publicly shared ChatGPT conversation containing two complete proofs of the value 1/12 [2]. The result, the antipodal mechanism, the total-sum threshold and the linear-dependence argument below are already in that conversation. Our purpose is expository: organise the proof in one table and express the count through the faces of one tetrahedron, simplifying the final averaging calculation. The author posted a version of the argument on MathOverflow on 1 October 2026 [7], found with Claude (Anthropic) but without having read that conversation.

Related exact computations in geometric probability include Wendel's centre-containment probabilities [8], Hall's acute-triangle probabilities for points in a ball [3], and Kabluchko's expected solid-angle sums of random inscribed simplices [4].

The proof has three steps. First, antipodal flips turn the twelve angle conditions into a placement rule for sixteen triple sums. Second, the sums occupy a corner of a 4×4 table, and counting its six possible shapes gives an identity relating successful tetrahedra to their short face sums. Third, one spherical-cap calculation and one integral give the two expectations needed to obtain 1/12.

2. The angle criterion

The example suggests using antipodes: replacing any chosen point by its negative preserves its uniform distribution. The 16 choices of signs form eight pairs under overall reflection, which preserves angles. We keep a fixed and let T_X denote the tetrahedron obtained by flipping the vertices in $X \subseteq \{b,c,d\}$. Write N for the number of these eight tetrahedra having four acute

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faces. Its value is 1 for our example; $\mathbb{E}N$ averages this count over random choices of the four points. If p is the desired probability, symmetry and linearity of expectation give

$$\mathbb{E}N = 8p. \tag{2}$$

The eight tetrahedra are dependent; this equality does not require their independence.

In the failed face above, $\|a+d-c\|^2 = 5/9$ and $-2/9 = (5/9-1)/2$. The angle test has a particularly useful form (Figure 1).

Lemma 1. *For unit vectors x, y, z,*

$$(x-z)\cdot(y-z) = \tfrac{1}{2}\big(\|x+y-z\|^2 - 1\big).$$

Thus the angle at z in xyz is obtuse precisely when $\|x+y-z\| < 1$.

Proof. Both sides expand to $x\cdot y - x\cdot z - y\cdot z + 1$. □

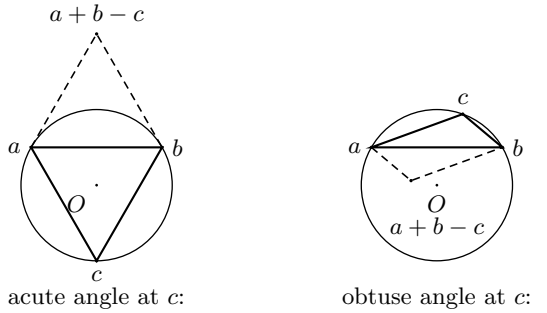


FIGURE 1. The fourth corner $x+y-z$ of the parallelogram on the sides at z lies outside the unit ball for an acute angle and inside for an obtuse one (the diagram uses a, b, c in place of x, y, z).

Our example has $\|a-c+d\|^2 = 5/9 < 1$. Call a vector sum *short* if its norm is less than 1. Before the flip it tests the obtuse angle at c; afterwards it is the vertex sum of $a(-c)d$ and tests no angle of that face. This change of role is what we need to track.

On each triple of antipodal pairs, the four triangles up to reflection have vertex sums

$$x+y+z, \qquad -x+y+z, \qquad x-y+z, \qquad x+y-z.$$

Flipping two vertices is, up to reflection, equivalent to flipping the third. There are four triples of pairs, hence sixteen labelled sums up to sign. For any one tetrahedron, four of these entries are its face vertex sums and the other twelve are its angle tests. Labels remain distinct even if vector sums coincide. We disregard sums of norm exactly 1, a null event justified in Section 6.

Lemma 2. *A tetrahedron has four acute faces if and only if every short entry in the sixteen-entry list is among its four face vertex sums.*

Proof. By Lemma 1, all twelve angles are acute exactly when all twelve angle-test sums have norm greater than 1. The only remaining entries are the four face sums. □

3. The table

A triangle belongs to two tetrahedra, obtained by either choice of the fourth vertex. For example, $a, -c, d$ belongs to T_c and T_{bc}, which flip one and two vertices respectively. Put the even choices $T_\varnothing, T_{bc}, T_{bd}, T_{cd}$ in rows and the odd choices T_b, T_c, T_d, T_{bcd} in columns, placing a triangle in the cell of its two containing tetrahedra.

To see why every cell holds exactly one triangle class, start with two examples. The pair $T_\varnothing, T_b$ differs only at b, so its common face is a, c, d. The pair $T_\varnothing, T_{bcd}$ appears to differ at three vertices, but reflecting T_{bcd} gives $(-a, b, c, d)$, revealing their common face b, c, d.

In general, an even and an odd choice differ at one or three of b, c, d. With one difference, the other three vertices give their unique common face. With three differences, reflection turns the pair into tetrahedra differing only at a, again giving a unique common face up to reflection.

Conversely, each triangle class belongs to one even and one odd tetrahedron. Choosing the fourth vertex from a b, c, d pair changes the parity by one flip. If the missing pair is a, reflecting to keep a fixed complements all three other signs, also switching even and odd. Thus every triangle class occupies exactly one cell, and all sixteen cells are filled.

Mark a cell if its vertex sum is short. A tetrahedron's line contains its four face sums; the twelve cells outside it are its angle tests. Thus Lemma 2 says that a tetrahedron succeeds exactly when its line contains every mark. The example in Table 1 has two marks in row T_{bc} and therefore $N = 1$.

The cell of T_{bc} and T_c has squared tetrahedron totals $20/9$ and $8/9$, adding to $28/9 < 4$; its face sum has squared norm $5/9 < 1$. The table also determines which cells are marked without testing each vector separately. Let x_i and y_j be the squared norms of the four-vertex totals of row i and column j. If their common face has vertex sum L and the remaining pair is $\pm A$, their totals are $L + A$ and $L - A$, up to sign. The parallelogram law gives

$$x_i + y_j = \|L + A\|^2 + \|L - A\|^2 = 2\|L\|^2 + 2. \tag{3}$$

Consequently a cell is marked exactly when $x_i + y_j < 4$. Sorting rows and columns by increasing squared totals makes the marks an upper-left corner: a marked cell forces every cell above it and to its left to be marked.

TABLE 1. One table for (1): entries are common-face vertex sums, up to sign; the margins give squared tetrahedron totals. A cell is marked when its two margins add to less than 4. The two marked entries (bold) have squared norm $5/9$. Row T_{bc} contains both; each of its twelve off-row entries is an acute-angle test.

		T_d	T_c	T_{bcd}	T_b
$x_i \backslash y_j$		$8/9$	$8/9$	$24/9$	$104/9$
T_{bc}	$20/9$	$\boldsymbol{-b-c+d}$	$\boldsymbol{a-c+d}$	$a-b-c$	$a-b+d$
T_{cd}	$36/9$	$a+b-d$	$a+b-c$	$a-c-d$	$b-c-d$
$T_\varnothing$	$44/9$	$a+b+c$	$a+b+d$	$b+c+d$	$a+c+d$
T_{bd}	$44/9$	$a+c-d$	$-b+c-d$	$a-b-d$	$a-b+c$

4. Restrictions on the marks

We now show that the marked corner is nonempty, has at most three marks in any line and has at most four marks in all. Averaging the four face sums gives the first two facts, and the parallelogram law gives the third. The nonemptiness argument is where dimension 3 enters.

For the example's T_{bc}, the four squared face sums are $5/9$, $5/9$, $53/9$ and $13/9$. Their average is $19/9 > 1$. The tetrahedron's total is $S = (-4, 0, 2)/3$, so this is $1 + \|S\|^2/2$. An average above 1 cannot force a short sum. To force one, we look for nonnegative weights totalling 1 that make the weighted average exactly 1; the following identity shows how to choose them.

For unit vertices $B_1, \ldots, B_4$ and total $S = \sum_i B_i$, the face sums are $S - B_i$. If $\beta_i \geq 0$ and $\sum_i \beta_i = 1$, then

$$\sum_i \beta_i \|S - B_i\|^2 = \|S\|^2 + 1 - 2S \cdot \sum_i \beta_i B_i. \tag{4}$$

The left side averages squared face sums; the right side shows that making the weighted vertex sum $S/2$ will make that average 1.

Lemma 3. *The marked corner is nonempty, has at most three marks in any line, and has at most four marks in total.*

Proof. Equal weights $\beta_i = 1/4$ in (4) give $1 + \|S\|^2/2 \geq 1$. Thus the four face sums cannot all be short, excluding a line of four marks.

For nonemptiness, half-weights give vertex sum $S/2$ but total weight 2. We must subtract weights totalling 1 whose vertex sum is zero. In the example, $3a - 2b - 9c + 6d = 0$ becomes a positive relation for T_{bc}; normalising gives $w = (3, 2, 9, 6)/20$. Subtract these from the half-weights:

	a	$-b$	$-c$	d	weight total	vertex sum
$\frac{1}{2}$ each	10/20	10/20	10/20	10/20	2	$S/2$
minus w	3/20	2/20	9/20	6/20	1	0
β	7/20	8/20	1/20	4/20	1	$S/2$

This subtraction corrects the total weight without changing the vertex sum, so (4) makes the squared-face average exactly 1. Indeed, with the squared face sums $5/9, 5/9, 53/9, 13/9$ opposite $a, -b, -c, d$,

$$\frac{7 \cdot 5 + 8 \cdot 5 + 1 \cdot 53 + 4 \cdot 13}{20 \cdot 9} = 1.$$

In general, four vectors in $\mathbb{R}^3$ have a nonzero linear relation. Flip vertices with negative coefficients; zero coefficients stay zero. Normalising the now nonnegative, not-all-zero coefficients produces one of the eight tetrahedra and weights $w_i \geq 0$ with $\sum_i w_i = 1$ and $\sum_i w_i B_i = 0$. Each satisfies

$$w_i = \Big\| \sum_{j \neq i} w_j B_j \Big\| \leq \sum_{j \neq i} w_j = 1 - w_i.$$

Thus $\beta_i = 1/2 - w_i$ are nonnegative, sum to 1 and have vertex sum $S/2$. Equation (4) gives average 1, so some face sum has norm at most 1, hence less than 1 under our null-set convention.

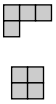
Finally, two distinct triangle classes on the same triple of pairs differ, after reflection if needed, by one flip. Their sums are $u + v$ and $u - v$ with $\|v\| = 1$, so their squared norms add to $2\|u\|^2 + 2 \geq 2$. They cannot both be short. There are four triples of pairs, giving at most four marks altogether. □

5. The counting identity

In Table 1, T_{bc} has two short face sums, T_c and T_d have one each, and the other five tetrahedra have none. Let k count the short face sums of one tetrahedron and let C count the tetrahedra with $k = 1$ or 2. The example has $C = 3$ and $N = 1$. The facts of Lemma 3 matter here: an empty table would give $N + C = 8$, and a fully marked row would give $N = 1$ and $C = 4$.

Lemma 4. *For every configuration outside the excluded null set, $N + C = 4$.*

Proof. By Lemma 3, the corner contains between one and four marks, with at most three per line. List its nonzero row lengths. Up to transposition, which changes neither count, there are six possibilities:

shape	row lengths	column lengths	N	C
	1	1	2	2
	2	1, 1	1	3
	3	1, 1, 1	1	3
	2, 1	2, 1	0	4
	3, 1	2, 1, 1	0	4
	2, 2	2, 2	0	4

Here N counts lines containing every mark, while C counts lines with one or two marks. For instance, in shape $(2, 1)$ no line contains every mark, but the two nonempty rows and two nonempty columns each count towards C. Every row of the displayed list gives $N + C = 4$. □

Taking expectations, using (2) and the equal distributions of the eight tetrahedra, gives

$$p = \tfrac{1}{2} - \Pr(k = 1 \text{ or } 2).$$

Two short faces give one pair, and three give three pairs. Writing the pair count as $\binom{k}{2}$, and recalling $k \le 3$, these counts encode precisely this event:

k	0	1	2	3
$\binom{k}{2}$	0	0	1	3
$k - \binom{k}{2}$	0	1	1	0.

In words, the number of short face sums minus their number of pairs is the indicator of $k = 1$ or 2. Therefore

$$p = \tfrac{1}{2} + \mathbb{E}\binom{k}{2} - \mathbb{E}k. \tag{5}$$

We now need only the probability of one short sum and of a pair sharing two vertices.

6. The cap calculation

The cap below height $-1/2$ occupies $1/4$ of the unit sphere. More generally, Archimedes' hat-box theorem makes a uniform point's height along any fixed axis uniform on $[-1, 1]$: a band of height h has area $2\pi h$. [1]

Fix a, b, and write $s = \|a + b\|$. Since

$$\|a + b + c\|^2 = s^2 + 1 + 2c \cdot (a + b),$$

the sum is short when the height of c along $(a + b)/s$ is below $-s/2$. Thus the cap has height $1 - s/2$ and probability

$$q = \frac{2 - s}{4}. \tag{6}$$

For example, $a \cdot b = -1/2$ gives $s = 1$ and $q = 1/4$; independent c, d both hit that fixed cap with probability $1/16$. Norm-one equality cases have probability zero: for $s > 0$, the squared norm above is uniform on an interval of positive length, and $s = 0$ itself has probability zero. Sign symmetry applies this argument to each of the sixteen labelled sums, justifying the earlier exclusion.

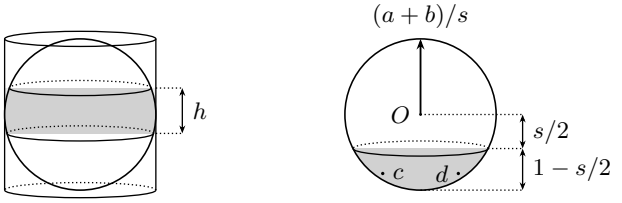


FIGURE 2. Equal-height sphere and cylinder bands have equal area (left). Both face sums $a + b + c$ and $a + b + d$ are short when c, d hit the same cap of height $1 - s/2$ (right).

Let $\alpha = \mathbb{E}q$ be the probability that one face sum is short. Two face sums share two vertices, so their probability of both being short is $\gamma = \mathbb{E}(q^2)$: fix a, b, multiply the two independent cap probabilities, then average. The shared random cap is why squaring must precede averaging. Four faces and six pairs give $\mathbb{E}k = 4\alpha$ and $\mathbb{E}\binom{k}{2} = 6\gamma$.

[1] At height z, the ring radius is $r = \sqrt{1 - z^2}$. The circumference $2\pi r$ times the sloping width per unit height $\sqrt{1 + (dr/dz)^2} = 1/r$ gives constant area density 2π. Integrating gives the band area, with pole endpoints understood by limits.

It is useful to average the hit–miss probability $q(1-q)$, whose factors remove the square root in s:

$$q(1-q) = \frac{(2-s)(2+s)}{16} = \frac{4-s^2}{16}, \qquad \mathbb{E}s^2 = 2 + 2\mathbb{E}(a \cdot b) = 2.$$

The last mean vanishes by the symmetry $b \leftrightarrow -b$. Hence $\alpha - \gamma = 1/8$, and (5) reduces to

$$p = \tfrac{1}{2} + 6\gamma - 4\alpha = 2\alpha - \tfrac{1}{4}.$$

Finally, $U = (1 + a \cdot b)/2$ is uniform on $[0, 1]$ and $s = 2\sqrt{U}$, so

$$\alpha = \int_0^1 \frac{1-\sqrt{u}}{2}\, du = \frac{1}{6}, \qquad p = 2 \cdot \frac{1}{6} - \frac{1}{4} = \boxed{\frac{1}{12}}.$$

As a check, α is also the probability that a specified angle of an inscribed random triangle is obtuse. Its three angles have equal distributions and at most one is obtuse, giving acute-triangle probability $1 - 3\alpha = 1/2$.

7. Remarks

The varying success count $N \in \{0, 1, 2\}$ is what prevents a constant-flip argument of the kind used for centre containment [8]. The earlier proof [2] counts $N = 4 - 2m + J$, where m is the number of short entries and J counts pairs on a common line. Each entry lies on two lines and two distinct entries share at most one line, so $\sum_t k_t = 2m$, $\sum_t \binom{k_t}{2} = J$ and $C = 2m - J$. Thus $N + C = 4$ is the same identity expressed through the short face sums of one tetrahedron; its advantage is the immediate reduction to two moments.

Dimension 3 enters in the nonzero linear relation among four vertices. For four orthonormal vectors in $\mathbb{R}^4$, every triple sum has squared norm 3: the table is empty and all eight tetrahedra have equilateral faces. The short-sum angle test is unchanged, but the nonemptiness argument fails.

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References

[1] A. Akopyan, H. Edelsbrunner and A. Nikitenko, *The beauty of random polytopes inscribed in the* 2*-sphere*, Experiment. Math. **34** (2025), no. 3, 327–341. doi:10.1080/10586458.2021.1980459.

[2] *Tetrahedron Acute Faces Proof*, ChatGPT-generated responses in a conversation publicly shared by Timothy Chow and linked in a MathOverflow comment on 14 August 2026. `https://chatgpt.com/share/6a7e8742-01e0-83ea-81de-533ea9dcb597`. Accessed 6 October 2026.

[3] G. R. Hall, *Acute triangles in the n-ball*, J. Appl. Probab. **19** (1982), no. 3, 712–715. doi:10.2307/3213533.

[4] Z. Kabluchko, *Angle sums of random simplices in dimensions* 3 *and* 4, Proc. Amer. Math. Soc. **148** (2020), no. 7, 3079–3086. doi:10.1090/proc/14934.

[5] Dan, *A tetrahedron's vertices are random points on a sphere. What is the probability that the tetrahedron's four faces are all acute triangles?*, MathOverflow question 498296 (2025), with answers and comments, `https://mathoverflow.net/q/498296`.

[6] Dan, the same question on Mathematics Stack Exchange, question 5083355 (July 2025), with answers by J. D'Aurizio and J. Ferry, `https://math.stackexchange.com/q/5083355`.

[7] V. Jandhyala, answer to MathOverflow question 498296, 1 October 2026, `https://mathoverflow.net/a/515661`.

[8] J. G. Wendel, *A problem in geometric probability*, Math. Scand. **11** (1962), 109–111. doi:10.7146/math.scand.a-10655.

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